AP CALCULUS AB

Unit 1 • Worksheet 1-6A

Worksheet 1-6A • Average Rate of Change

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

What is the average rate of change of $f(x) = x^3 - 2x$ on $[-1, 2]$?

(A) 3

(B) $\frac{5}{3}$

(C) 1

(D) -1

Question 2

NO CALCULATOR

For $f(x) = x^2 + x$, which is $\dfrac{f(2+h) - f(2)}{h}$ for $h \neq 0$?

(A) 5

(B) $5 + h$

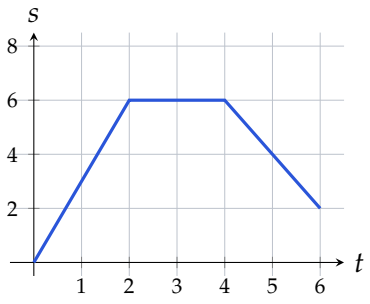
(C) $4 + h$

(D) $h^2 + 5h$

Question 3

NO CALCULATOR

The position (m) of a particle is graphed. What is its average velocity on $[2, 6]$?



(A) -1 m/s

(B) -2 m/s

(C) 1 m/s

(D) 0 m/s

Question 4

NO CALCULATOR

$V(t)$ is a volume in liters, t in minutes. Units of $\frac{V(8)-V(2)}{6}$?

(A) liters
(B) minutes per liter
(C) liters per minute
(D) liters per minute per minute

Question 5 (Free response)

NO CALCULATOR

t (hours)	0	4	8	12	16
$T(t)$ (°F)	52	48	57	71	66

(a) Find and interpret the average rate of change of T on $[4, 12]$.
(b) On which interval of length 4 did T increase fastest on average?
(c) For $s(t) = t^3$, simplify $\frac{s(1+h)-s(1)}{h}$ and evaluate it for $h = 0.1$.